

Chapter 4. HIERARCHICAL STRUCTURE OF A COMPUTER SYSTEM

MEMORY

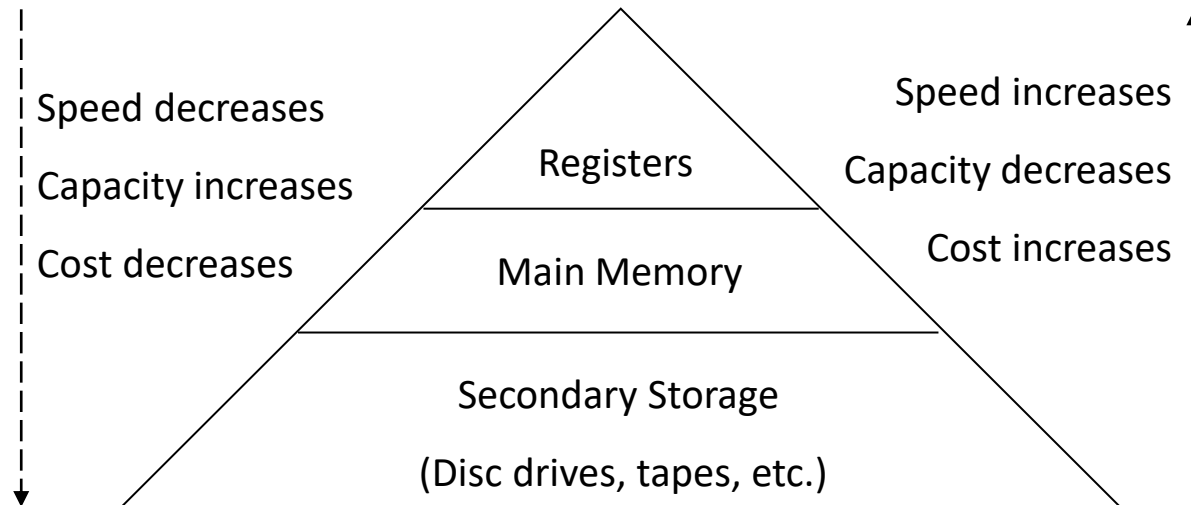
ROM – Read Only Memory

RAM – Random Access Memory

Registers in a CPU

MEMORY

So far, we have used flip-flops and registers to load and process binary data. But the number of registers in a computer system is no more than a few hundreds. The number of registers is limited due to cost, energy consumption, and the volume occupied in the CPU. However, programs we need to run on a computer system is larger than we can store in these registers. Memories in computer systems have a hierarchy.



Memory Types

Basic semi-conductor memory types are RAM (Random Access Memory) and ROM (Read-Only Memory). While, RAM allows reading and writing, ROM allows only reading. Data stored in RAM is volatile. So, when the power is lost, the data stored in the RAM gets lost.

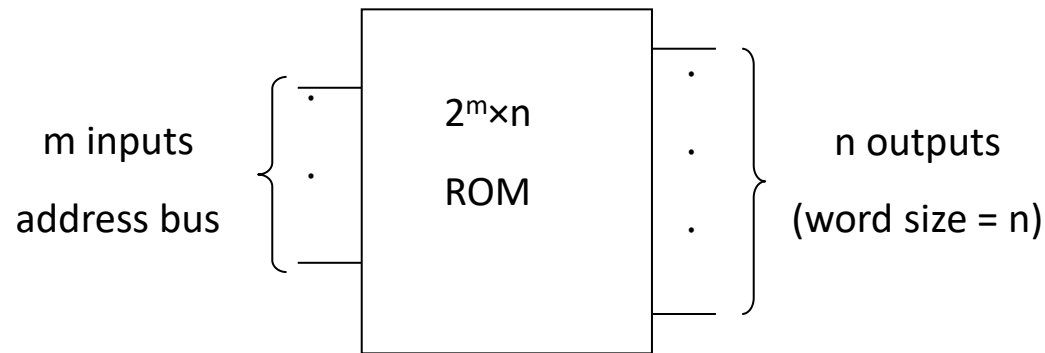
The data stored in these memory components is called “byte” or “word”. Byte represents 8-bit data. But number of bits in a “word” depends on the architecture.

In terms of CPU, word is the number of bits that a CPU can process at a time. But in terms of memory, word is the number of bits that is stored in each memory location.

A word size can be 1, 2, 4, 8, ... bytes depending on the architecture.

ROM – Read Only Memory

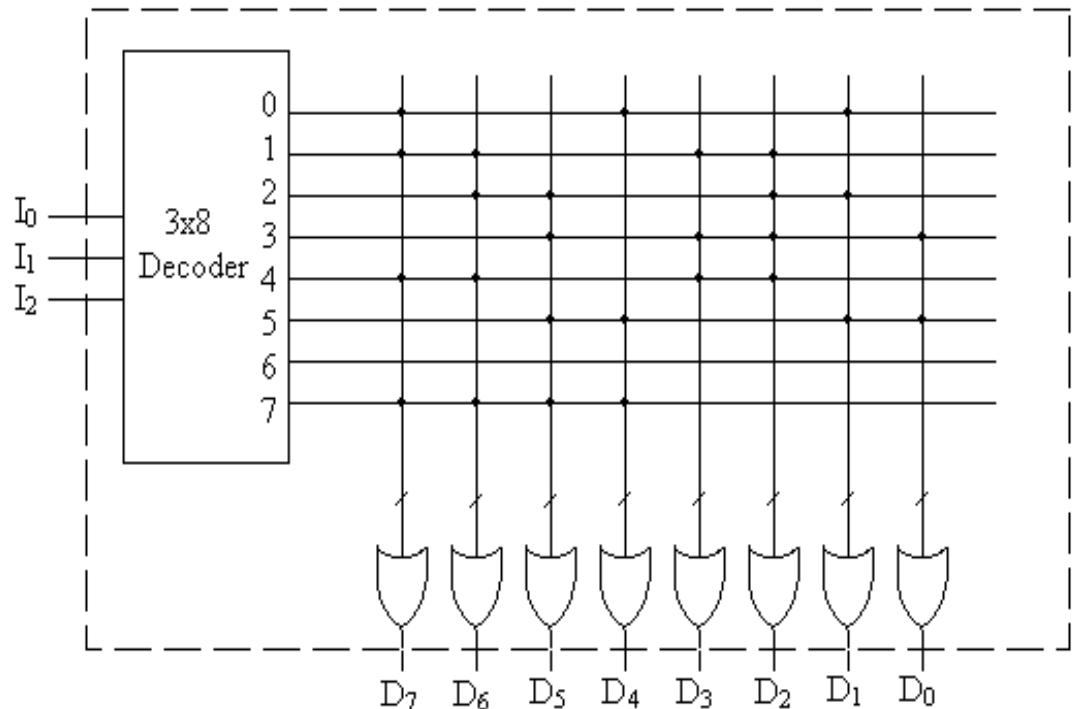
A ROM can be considered as a combinational circuit that consists of a decoder and OR gates.



ROM – Read Only Memory

Example: We can design an 8×8 ROM that stores the data given in the table as given below.

Inputs			Outputs
I ₂	I ₁	I ₀	D ₇D ₀
0	0	0	10010010
0	0	1	11001100
0	1	0	01100110
0	1	1	00101101
1	0	0	11001100
1	0	1	00110011
1	1	0	00000000
1	1	1	11110000



The given ROM has 8 memory locations. Therefore, we need 3 bits to address these locations. ($2^3=8$)

The number of locations depends on the number of address inputs. But the number of bits in each location (word size) does not depend on address inputs.

ROM Types

ROM

A.K.A. Mask ROM. Stores data permanently. Programmed during manufacturing.

PROM

Programmable ROM. Programmed after manufacturing.

EPROM

Erasable PROM. Can be erased by exposing to strong ultraviolet light source.

EEPROM

Electrically Erasable PROM. Has to be completely erased before rewriting.

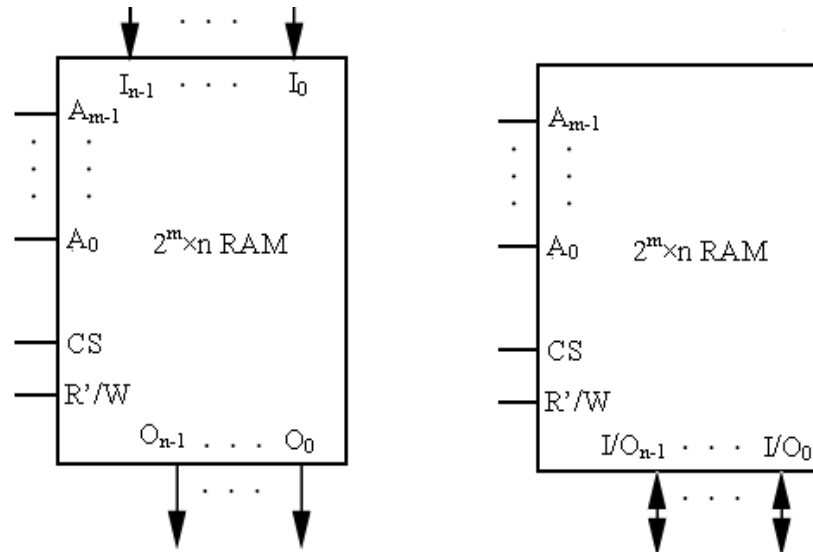
FLASH

Based on EEPROM. No need to erase the entire data, can be erased, written, and read in blocks. Used in USB flash drives.

RAM – Random Access Memory

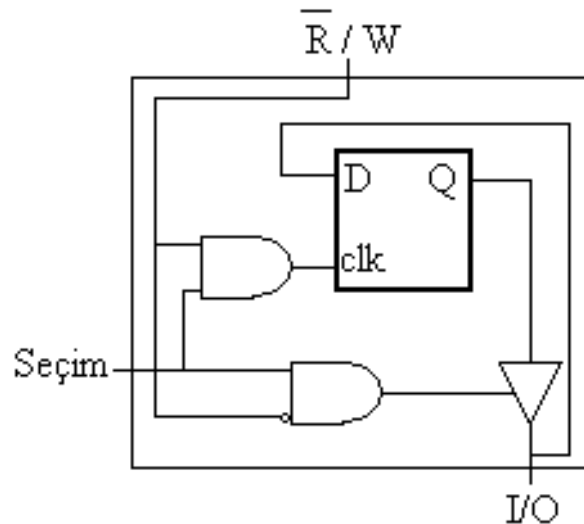
Data is volatile. You can access to any location at same speed. We can say the same thing for ROM. However, it is not the same in Sequential Access Memories.

A RAM with m address inputs has 2^m memory locations. Addresses of the locations are between 0 and (2^m-1) . Each location stores n -bit data which we call a memory word. We call such a memory as a $2^m \times n$ memory.



RAM Cell

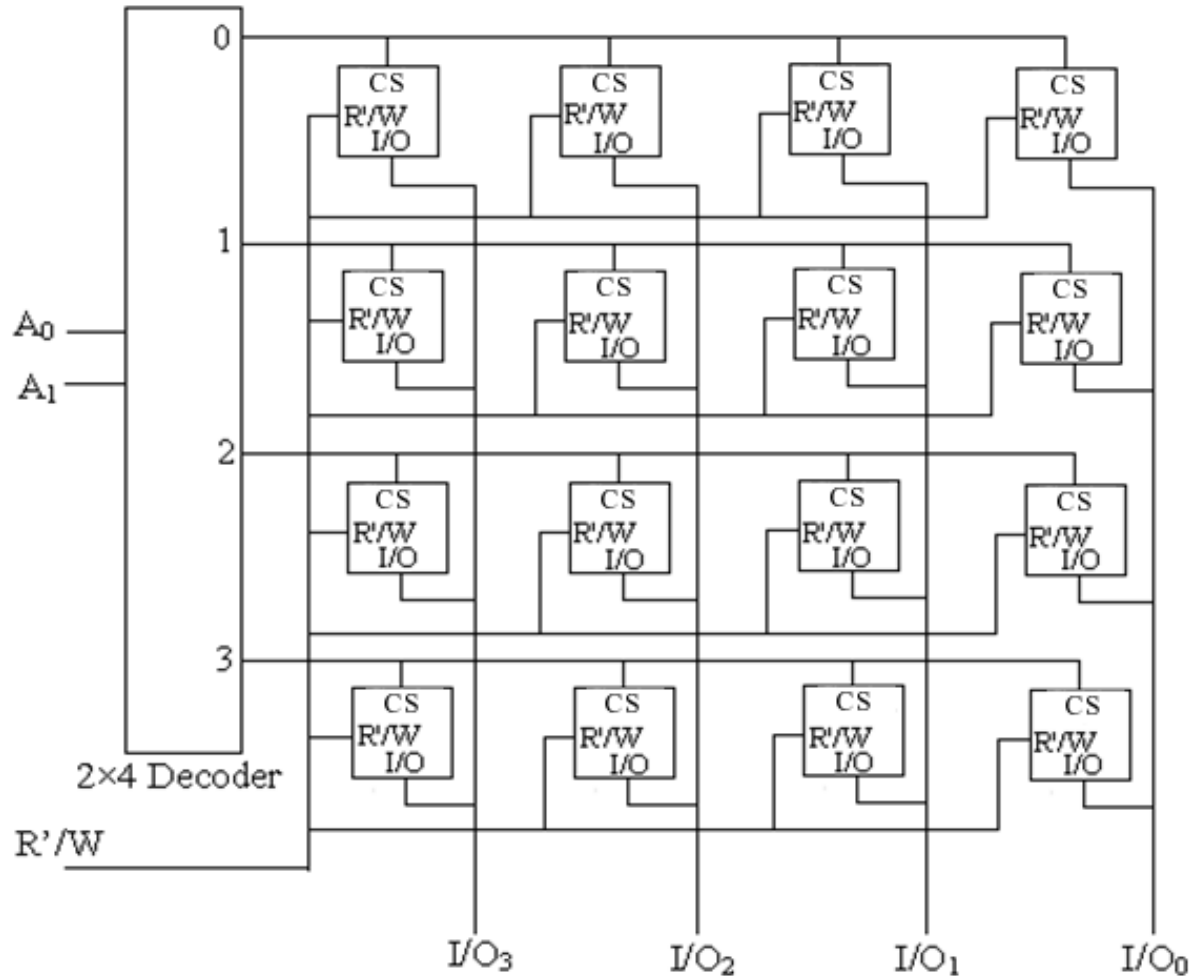
A RAM consists of RAM cells that stores 1-bit data. A RAM cell can be designed as given below.



RAM Chip

We can design a RAM chip using a decoder and RAM cells as given below.

A 4×4 RAM chip can be designed as given below.



RAM Organization

We can access memory locations of a low-capacity memory with a single decoder (row-based access). However, we need to use two decoders to access memory locations of a high-capacity memory (row/column based access).

For example, let's consider that we have a 1K word memory (a memory with 1K locations) and we need to access the location 321 (0101000001_2). We can organize the memory in two ways.

1st Way: To address the whole memory, we need 10 bits ($2^{10}=1K$). Therefore, we need to use a 10×1024 decoder.

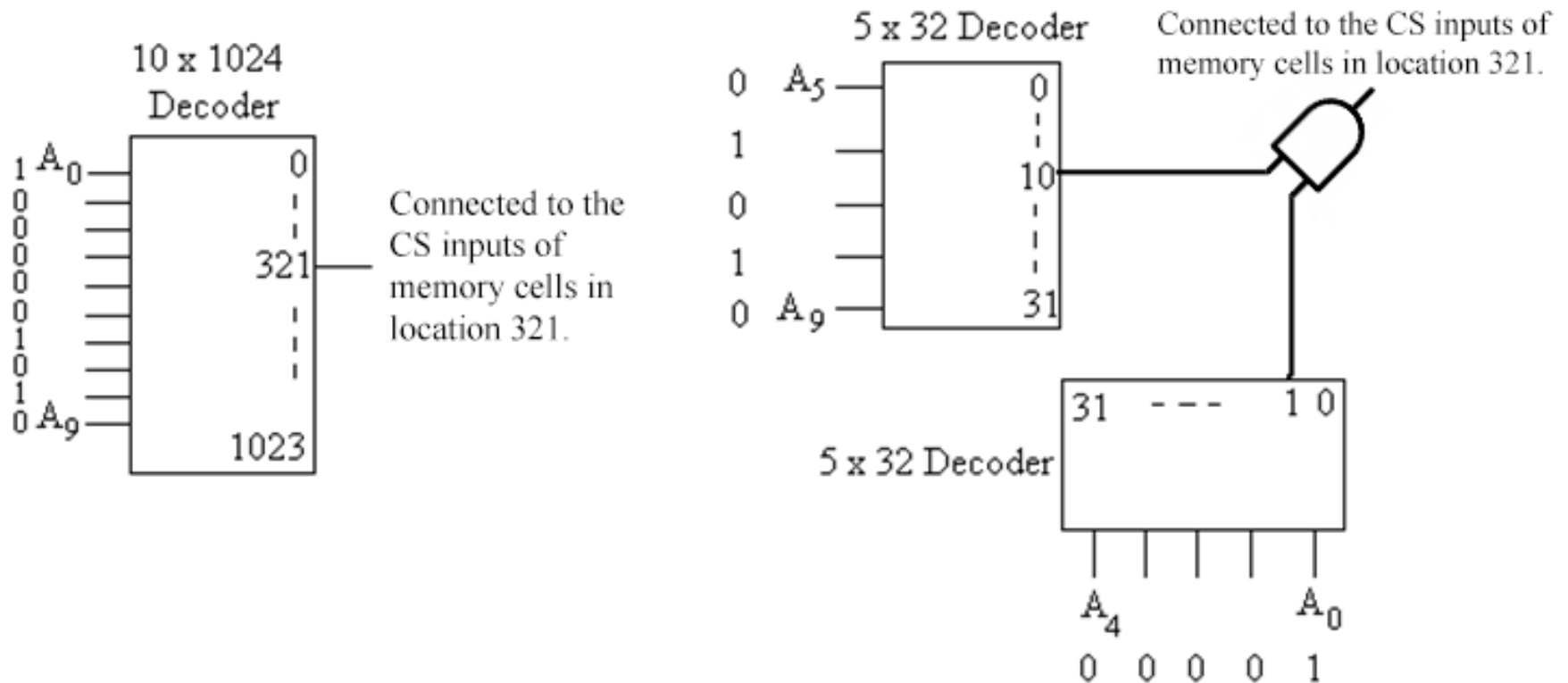
2nd Way: To address the whole memory, we need two 5×32 decoders ($2^5 \times 2^5 = 2^{10}$). We split the address into two parts, and then input these two parts to each decoders.

Row Column

01010 00001

$(10)_{10}$ $(1)_{10}$

RAM Organization



1st Way

2nd Way

RAM Types

There are two types of RAMs: Static RAM (**SRAM**), and Dynamic RAM (**DRAM**)

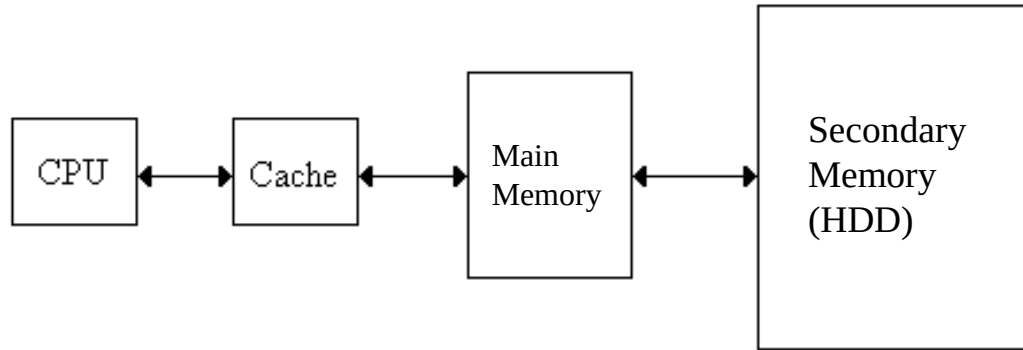
We can access data in an SRAM faster than a DRAM. But in an SRAM, each memory cell occupies more physical space than a memory cell in a DRAM.

An SRAM cell consists of 6 transistors whereas a DRAM cell consists of a transistor and a capacitor. But, since DRAMs have capacitors, they need to be refreshed periodically (we need to write the same data again).

DRAMs has a lower cost. We use DRAMs as main memories and SRAMs as cache memory. We transfer a certain part of the program from main memory to the cache, so that CPU can access data faster. We use both types of memories in order to balance performance and cost.

Memory Hierarchy

The memory hierarchy in a computer system is as follows.



Registers in a CPU

Since the CPU communicates with memory and I/O devices, it needs the registers given below.

PC (Program Counter)

IR (Instruction Register)

DR (Data Register)

AR (Address Register)

ACC (Accumulator)

INR (INput Register)

OUTR (OUTput Register)

TR (Temporary Register)

CCR (Condition Code Register)

IX (Index Register)

SP (Stack Pointer)